

Linear Algebra
Assignment – 01

by

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Question-03

$$\left. \begin{aligned} 2x - y + 3z - w &= -1.5 \\ -x + 4y - 2z + w &= 11 \\ 3x + 2y - z - 2w &= 2 \\ -2x - y + 4z + 3w &= 7 \end{aligned} \right\}$$

Converting into augmented matrix

$$\left[\begin{array}{cccc|c} 2 & -1 & 3 & -1 & -1.5 \\ -1 & 4 & -2 & 1 & 11 \\ 3 & 2 & -1 & -2 & 2 \\ -2 & -1 & 4 & 3 & 7 \end{array} \right]$$

Making first pivot

$$\left[\begin{array}{cccc|c} 1 & -0.5 & 1.5 & -0.5 & -0.75 \\ -1 & 4 & -2 & 1 & 11 \\ 3 & 2 & -1 & -2 & 2 \\ -2 & -1 & 4 & 3 & 7 \end{array} \right] \quad \text{--- } \frac{1}{2} R_1$$

Making Zero's under first pivot

$$\left[\begin{array}{cccc|c} 1 & -0.5 & 1.5 & -0.5 & -0.75 \\ 0 & 3.5 & -0.5 & 0.5 & 10.25 \\ 0 & 3.5 & -5.5 & -0.5 & 4.25 \\ 0 & -2 & 7 & 2 & 5.5 \end{array} \right] \quad \begin{array}{l} \text{--- } R_2 + R_1 \\ \text{--- } R_3 - 3R_1 \\ \text{--- } R_4 + 2R_1 \end{array}$$

Making second pivot

$$\left[\begin{array}{cccc|c} 1 & -0.5 & 1.5 & -0.5 & -0.75 \\ 0 & 1 & -0.143 & 0.143 & 2.929 \\ 0 & 3.5 & -5.5 & -0.5 & 4.25 \\ 0 & -2 & 7 & 2 & 5.5 \end{array} \right] \quad \text{--- } \frac{1}{3.5} R_2$$

Making Zeros in 2nd pivot column

$$\left[\begin{array}{cccc|c} 1 & 0 & 1.429 & -0.429 & 0.714 \\ 0 & 1 & -0.143 & 0.143 & 2.929 \\ 0 & 0 & -5 & -1 & -6 \\ 0 & 0 & 6.714 & 2.286 & 11.357 \end{array} \right] \begin{array}{l} R_1 + \frac{1}{2} R_2 \\ \\ R_3 - 3.5 R_2 \\ R_4 + 2 R_2 \end{array}$$

Making 3rd pivot

$$\left[\begin{array}{cccc|c} 1 & 0 & 1.429 & -0.429 & 0.714 \\ 0 & 1 & -0.143 & 0.143 & 2.929 \\ 0 & 0 & 1 & 0.2 & 1.2 \\ 0 & 0 & 6.714 & 2.286 & 11.357 \end{array} \right] -\frac{1}{5} R_3$$

Making Zeros in 3rd pivot column

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -0.714 & -1 \\ 0 & 1 & 0 & 0.171 & 3.1 \\ 0 & 0 & 1 & 0.2 & 1.2 \\ 0 & 0 & 0 & 0.943 & 3.3 \end{array} \right] \begin{array}{l} R_1 - 1.429 R_3 \\ R_2 + 0.143 R_3 \\ \\ R_4 - 6.714 R_3 \end{array}$$

Making 4th pivot

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -0.714 & -1 \\ 0 & 1 & 0 & 0.171 & 3.1 \\ 0 & 0 & 1 & 0.2 & 1.2 \\ 0 & 0 & 0 & 1 & 3.5 \end{array} \right] \frac{1}{0.943} R_4$$

Making Zeros in 4th pivot column

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1.5 \\ 0 & 1 & 0 & 0 & 2.5 \\ 0 & 0 & 1 & 0 & 0.5 \\ 0 & 0 & 0 & 1 & 3.5 \end{array} \right]$$

This is reduced Echelon form

from REF

$$x = 1.5, \quad y = 2.5, \quad z = 0.5, \quad w = 3.5$$

This system is a unique solution

Interpretation

There exist one precise allocation of 4 instruments that satisfies all audit constraints simultaneously.

The portfolio manager can assign

- 1.5 units in equity stocks
- 2.5 units in government bonds
- 0.5 units in commodity futures
- 3.5 units in REITS

Question - 02

$$\begin{cases} x + y + 7z = -7 \\ 2x + 3y + 17z = -16 \\ x + 2y + (a^2 + 1)z = 3a \end{cases}$$

In augmented matrix form

$$\left[\begin{array}{ccc|c} 1 & 1 & 7 & -7 \\ 2 & 3 & 17 & -16 \\ 1 & 2 & (a^2 + 1) & 3a \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 0 & 1 & (a^2 - 6) & 3a + 7 \end{array} \right] \quad \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 7 & -7 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & (a^2-9) & 3a+9 \end{array} \right] R_3 - R_2$$

for z from last Row

$$(a^2-9)z = 3a+9 \quad \because a^2-9 = (a+3)(a-3)$$

$$(a-3)(a+3)z = 3(a+3) \quad \text{--- (1)}$$

CASE STUDY

CASE I: $a \neq \pm 3$

when $a \pm 3 \neq 0$

We can

$$(a-3)z = \frac{3(a+3)}{(a+3)}$$

$$z = \frac{3}{a-3}$$

Now solving for others put z in 2nd equation

$$y + 3z = -2$$

$$y + 3\left(\frac{3}{a-3}\right) = -2$$

$$y = -2 - \frac{9}{a-3}$$

Now, find x using values.

$$x + y + 7z = -7$$

$$x = -7 - y - 7z$$

$$x = -7 - \left(-2 - \frac{9}{a-3}\right) - \frac{21}{a-3}$$

$$x = -7 + 2 + \frac{9}{a-3} - \frac{21}{a-3}$$

$$x = -5 - \frac{12}{a-3}$$

Hence 1 unique solution

$$\left(-5 - \frac{12}{a-3}, -2 - \frac{9}{a-3}, \frac{3}{a-3}\right)$$

CASE II : $a=3$

put $a=3$ in eq — (A)

$$(a^2-9)z = 3a+9$$

$$(9-9)z = 9+9$$

$$0 \cdot z = 18$$

$$\boxed{0 = 18}$$

This is impossible. (No solution)

CASE III : $a=-3$

put $a=-3$ in eq — (A)

$$(9-9)z = -9+9$$

$$0z = 0$$

This is always true, hence z becomes a free variable.

We put $z=t$

put $z=t$ in eq — ②

$$y + 3z = -2$$

$$\boxed{y = -2 - 3t}$$

put $z=t$ in eq — ①

$$x = -7 - y - 7z$$

$$x = -7 - (-2 - 3t) - 7t$$

$$\boxed{x = -5 - 4t}$$

for when t is any Real number

for values

$$x = -5 - 4t$$

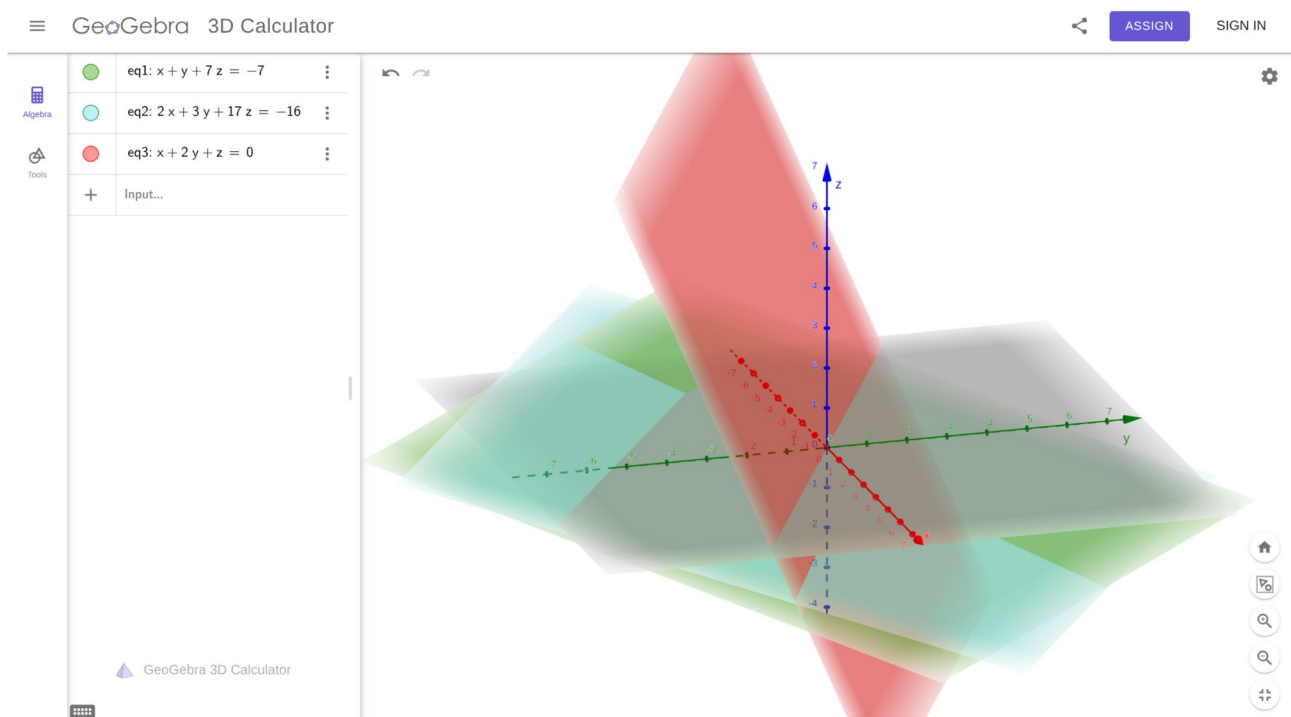
$$y = -2 - 3t$$

$$z = t$$

there lies infinite Solutions

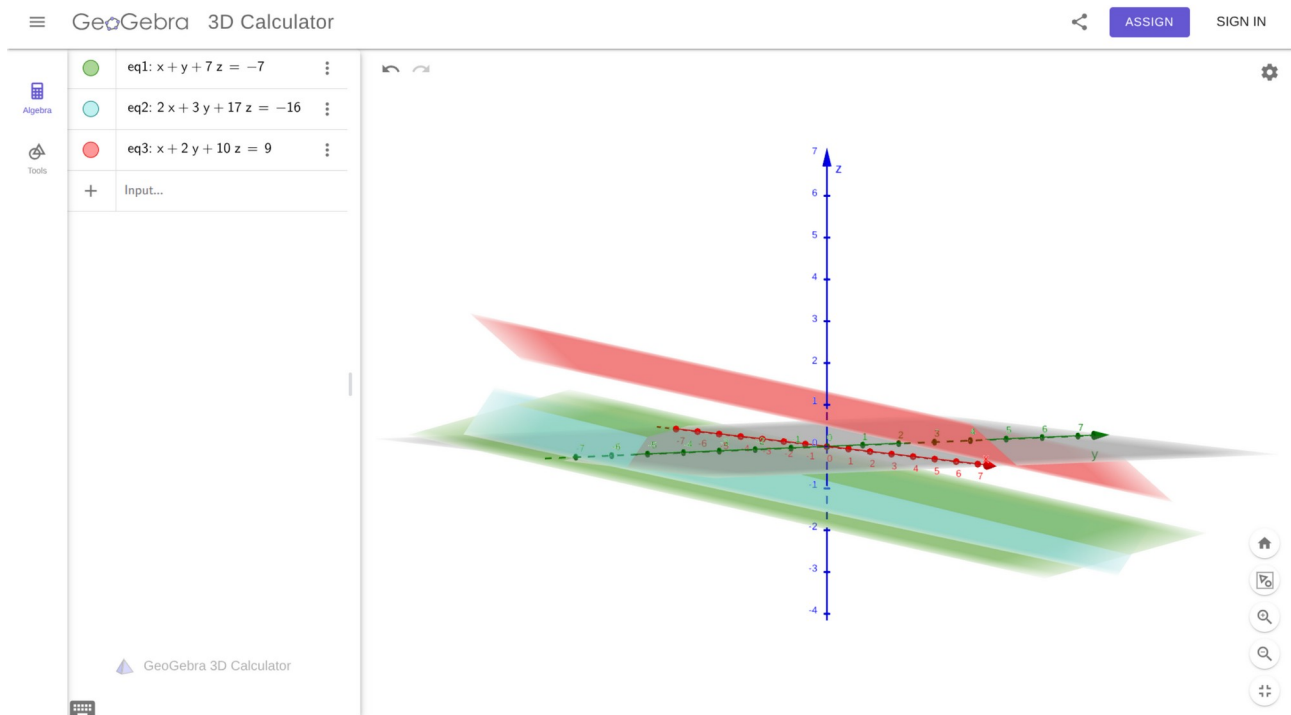
Case I: $a = 0$

Unique Solution



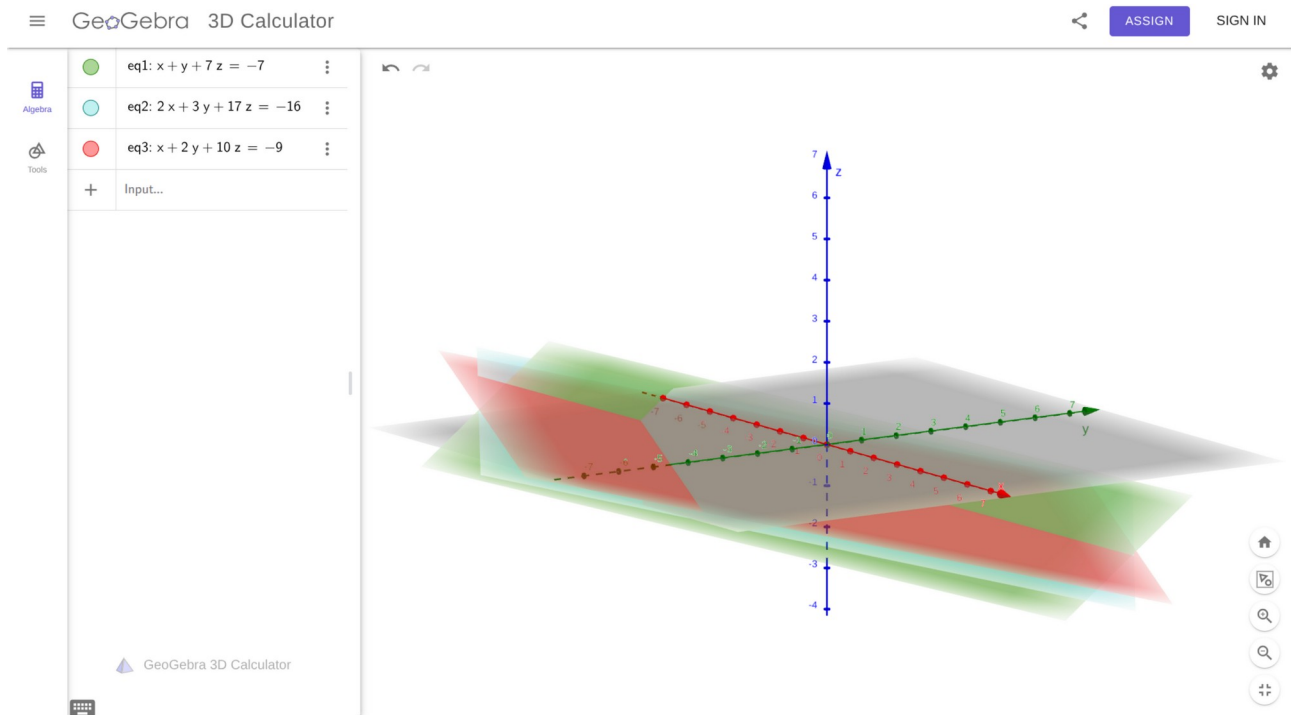
Case II: $a = 3$

No Solution



Case III: $a = -3$

Infinitely Many Solutions



Question - 01

UNIQUE SOLUTION

For a system to have to be unique all planes must intersect at exactly one point. This point is the only solution

Geometric Interpretation
Point

Example

$$\begin{cases} x + y + z = 6 \\ 2x - y + z = 3 \\ x + 2y - z = 3 \end{cases}$$

INFINITELY MANY SOLUTIONS

All the planes intersect in such a way that there are infinitely many solutions

- Either all planes intersect along a line, or
- All planes are lying on each other

Geometric Interpretation

Plane (or point if partially overlapping)

Example

$$\begin{cases} x + y + z = 3 \\ 2x + 2y + 2z = 6 \\ x + y + z = 3 \end{cases}$$

NO SOLUTION (INCONSISTENT SYSTEM)

The planes do not intersect at any point, they are parallel and distinct.

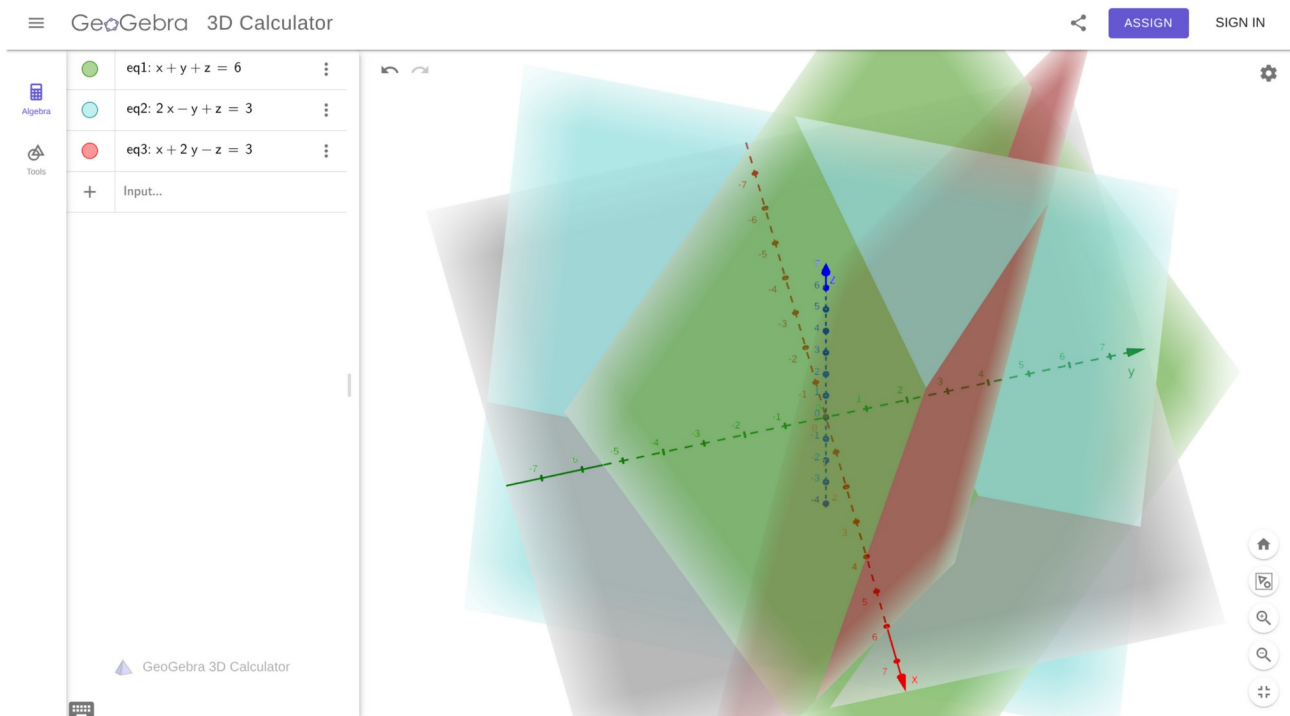
Geometric Interpretation

Parallel planes (No Intersection)

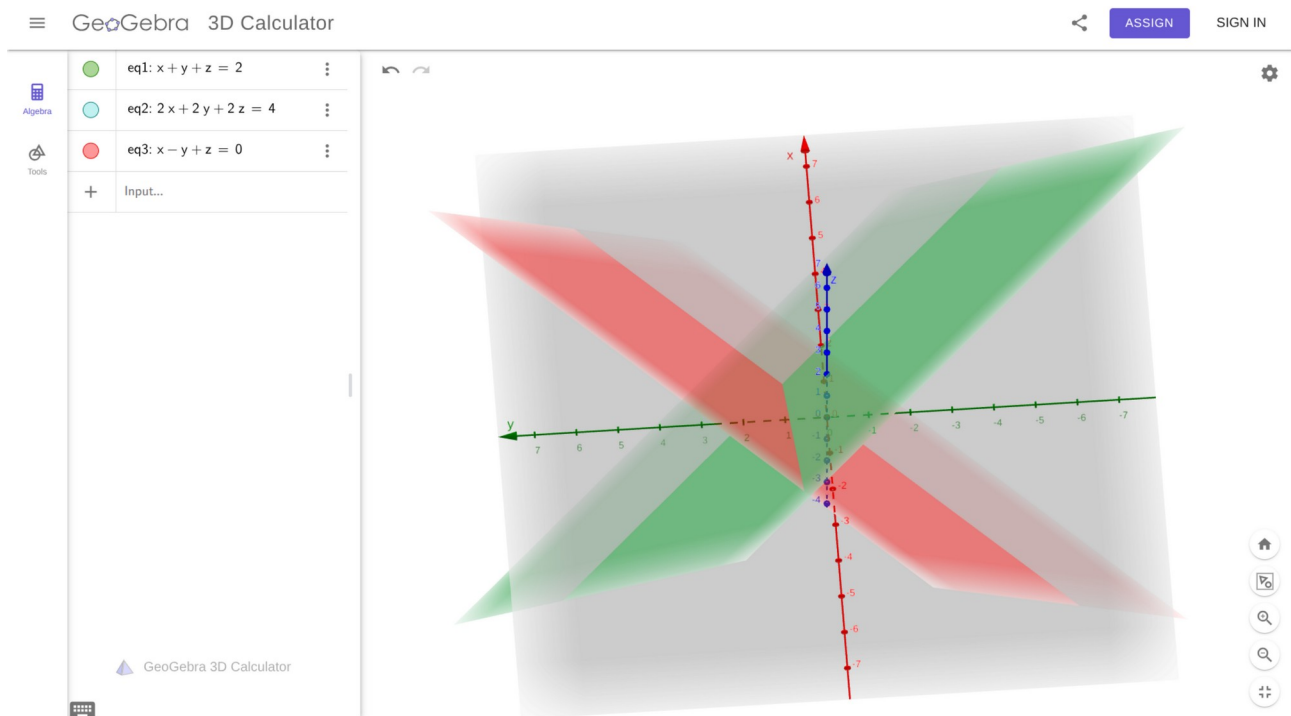
Example

$$\begin{cases} x + y + z = 1 \\ x + y + z = 3 \\ x - y + z = 2 \end{cases}$$

Case: Unique Solution



Case: Infinitely Many Solutions



Case: No Solution

